

EAS 6134: Inverse Methods in EAS

An introduction to inverse theory

12 January 2026

EAS 6134

Lecture 1: 1/12/2026

- Instructor: Samer Naif (snaif@eas.gatech.edu)
- Class website is now live on Canvas
- A brief intro to course topic
- Syllabus review
- Homework #1 due on Wednesday (!?)

Prerequisites

- Linear Algebra
- Partial Differential Equations
- Statistics and/or Data Analysis
- Computer coding (Matlab*, Python, Julia)

Sage advice: Inverse Theory is a challenging topic, including for your instructor. To succeed in this course, you need to stay ahead with the reading.

Come to lectures prepared. Life will be much more stressful if you do the reading after the fact to catch up on what we covered.

What is a Scientific Inverse Problem?

Process of uncovering the causal factors producing any set of observations

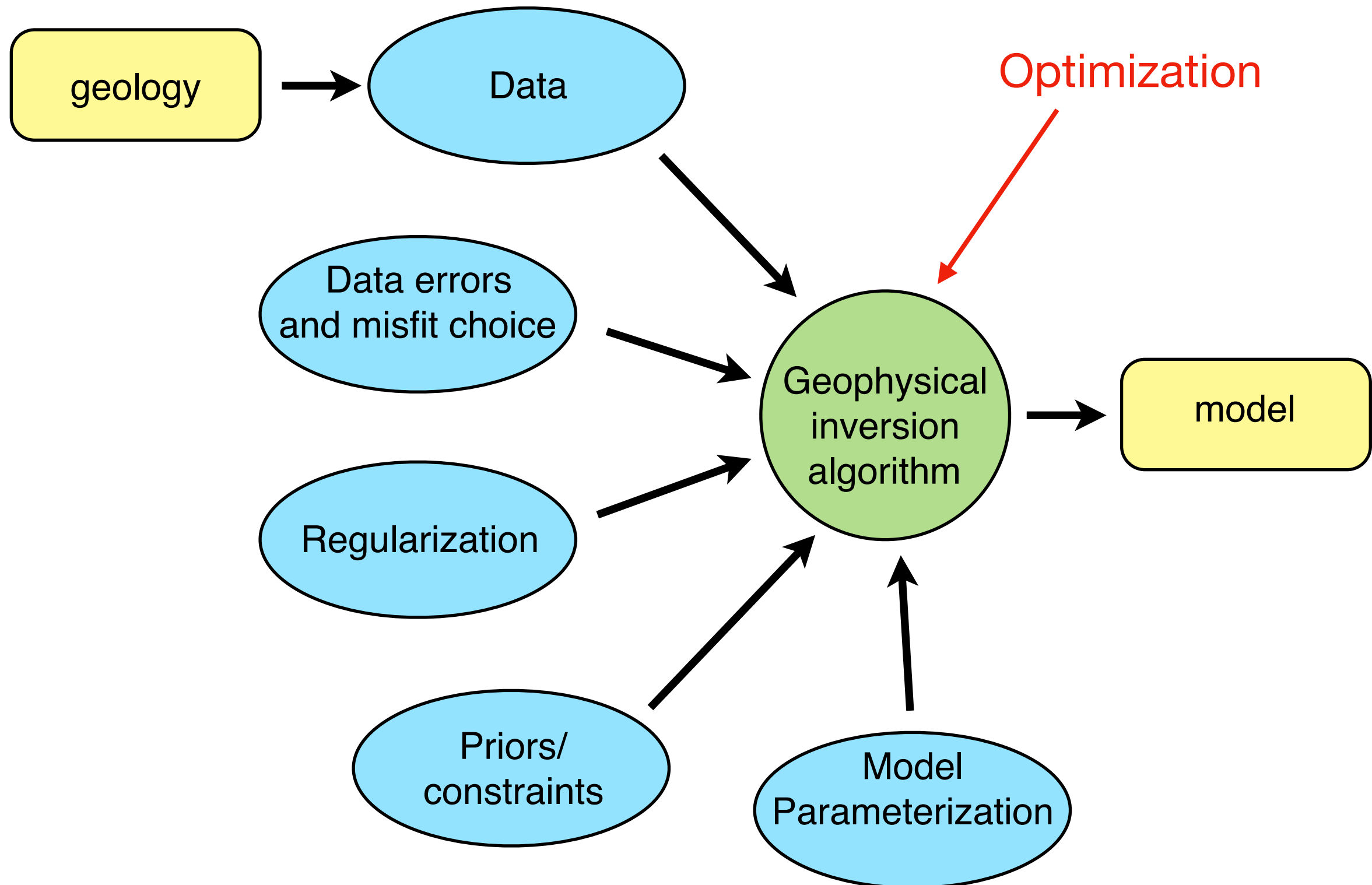
For example:

- earthquake source in seismology
- density of Earth from gravity measurements
- magnetization in Earth's lithosphere
- electrical conductivity or seismic velocity in Earth's mantle
- geomagnetic field at surface of Earth's core
- subglacial bed roughness in cryosphere

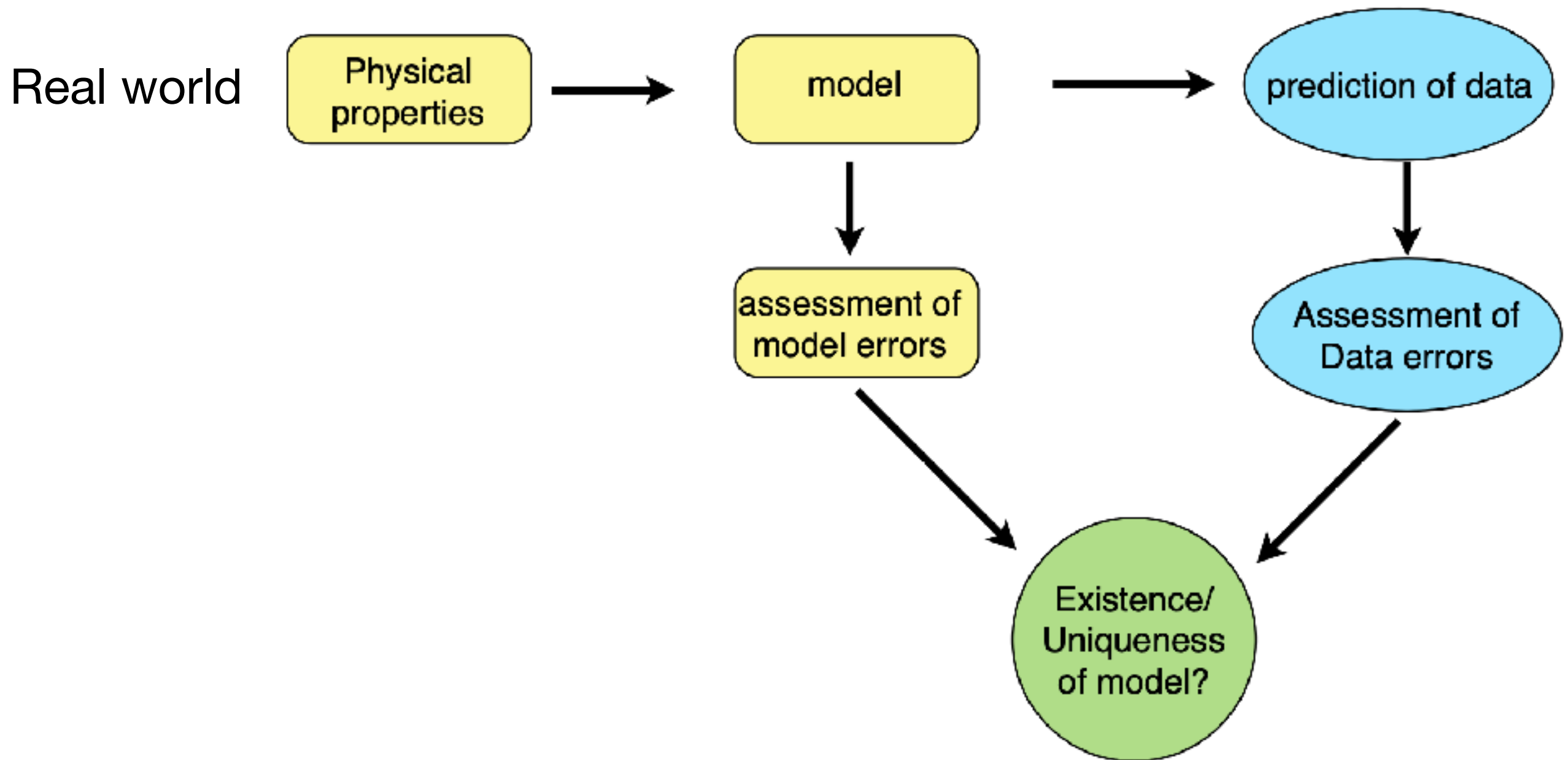
Applications in system identification, optics, radar, acoustics, communication theory, signal processing, medical imaging, computer vision, geophysics, oceanography, astronomy, remote sensing, numerical differentiation, natural language processing, machine learning, and more...

Common factor is that we cannot **directly** observe the thing we are interested in knowing.

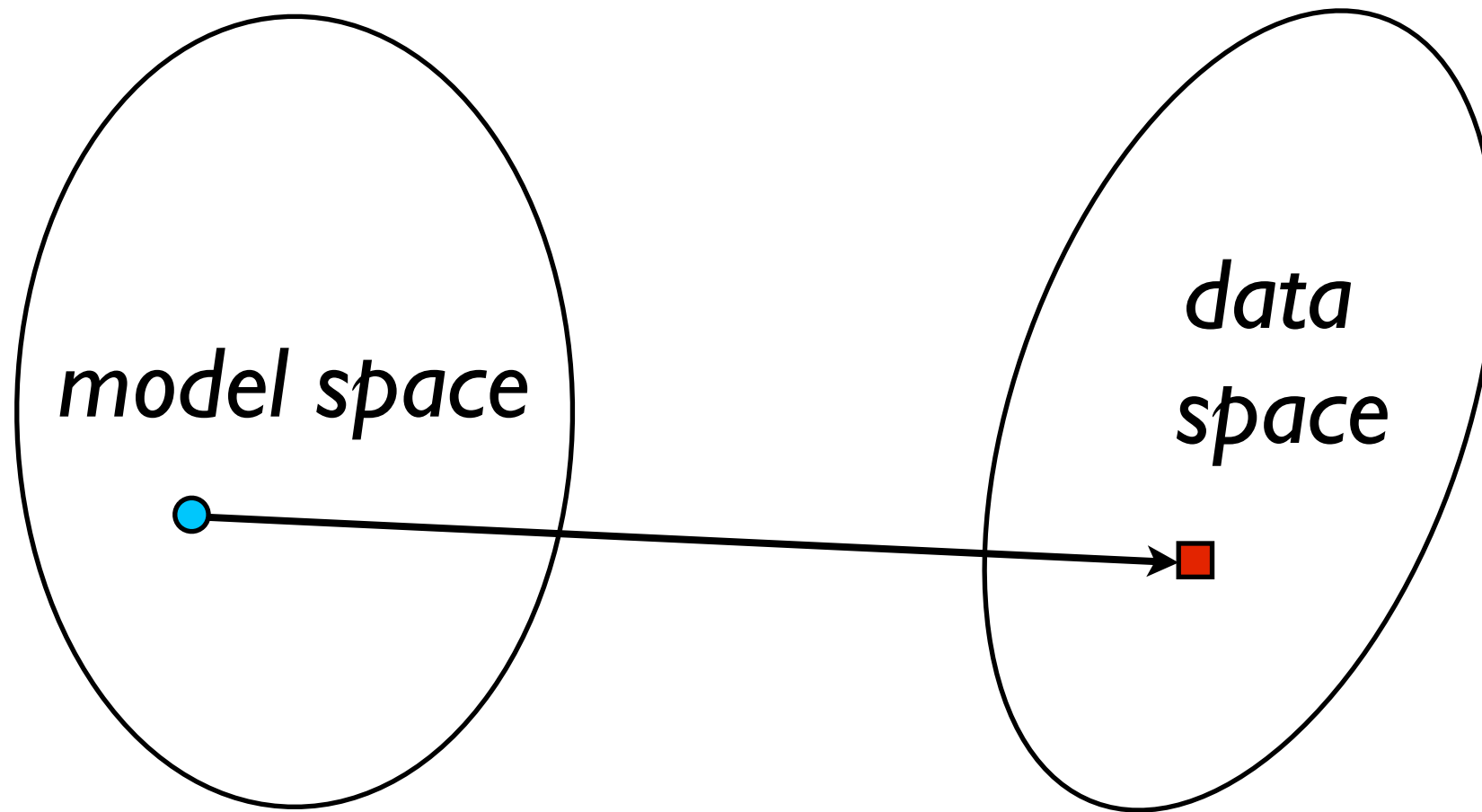
Geophysical inversion for model construction depends on much more than the data alone:



A geophysical forward model:



Forward modeling:



$$\hat{\mathbf{d}} = f(\mathbf{x}, \mathbf{m})$$

Some forward functional f

$$\mathbf{m} = (m_1, m_2, \dots, m_N)$$

Model parameters (layers, blocks, ...)

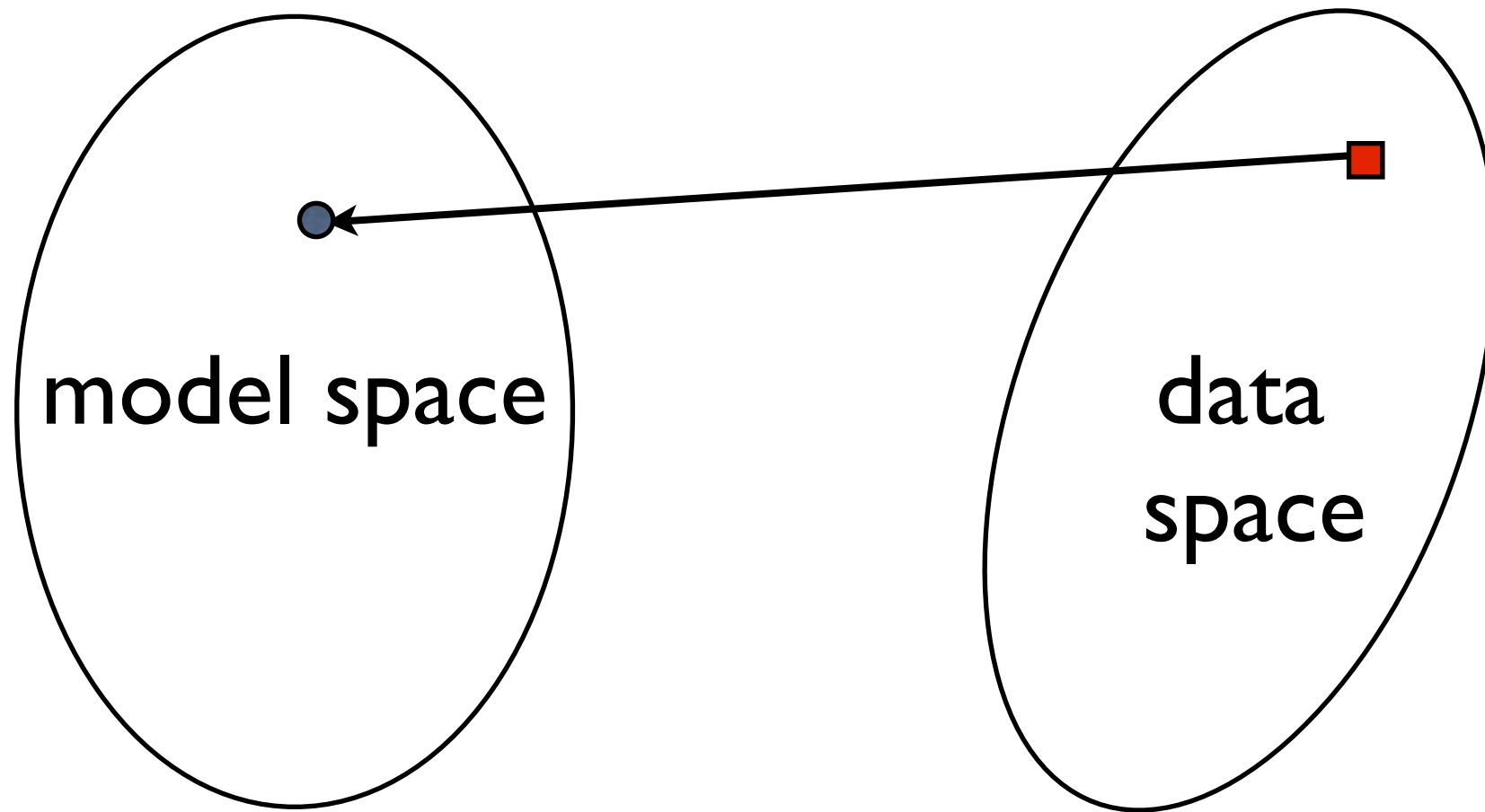
$$\mathbf{x} = (x_1, x_2, x_3, \dots, x_{kM})$$

Independent variables (freqs., locations, ...)

$$\hat{\mathbf{d}} = (\hat{d}_1, \hat{d}_2, \hat{d}_3, \dots, \hat{d}_M)$$

Predicted data (gravity, magnetic, electric, ...)

Inverse modeling:



Given real (observed) data
with errors
find an

$$\mathbf{d} = (d_1, d_2, d_3, \dots, d_M)$$

$$\sigma = (\sigma_1, \sigma_2, \dots, \sigma_M)$$

m

Gravity

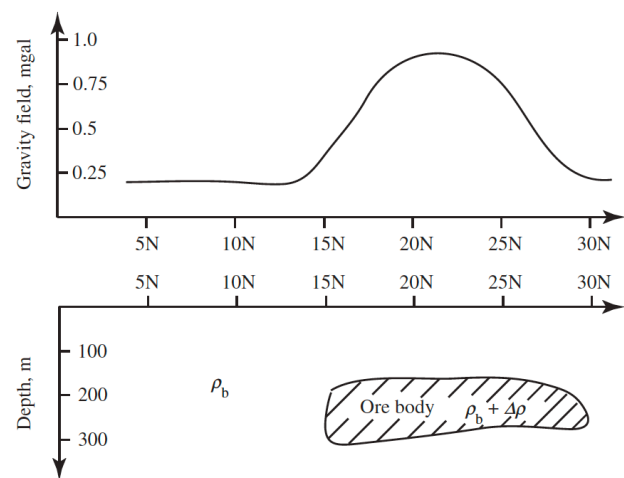


Figure 1.3

Anomalous gravity field profile over a massive ore body.

Seismic Survey

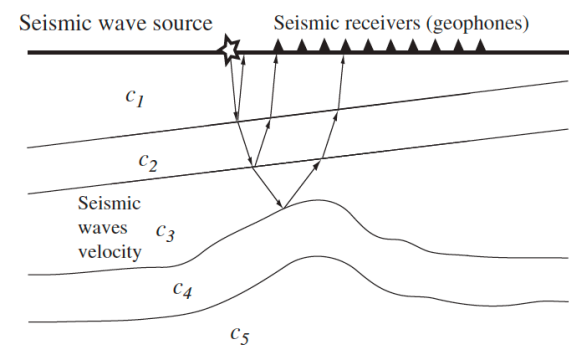


Figure 1.9

A typical reflection seismic survey.

Resistivity Survey

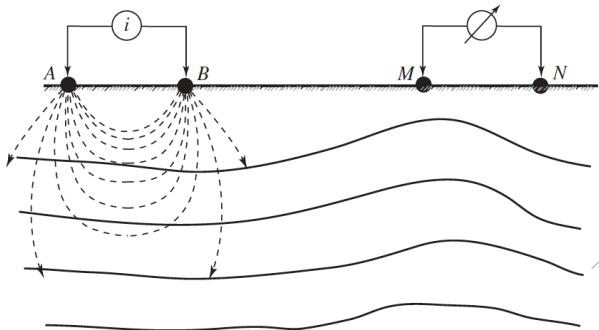


Figure 1.4

Current AB and receiver MN electrodes configuration in the typical DC geophysical survey.

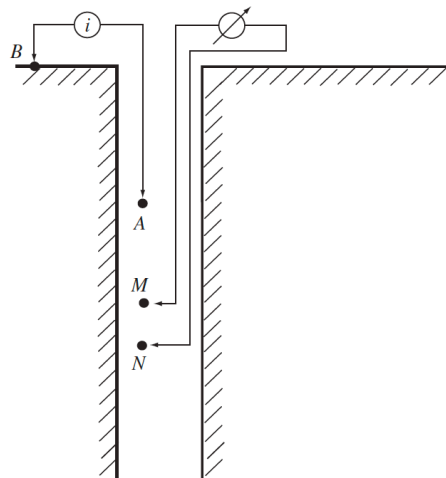


Figure 1.5

Current AB and receiver MN electrodes configuration in the resistivity well-logging.

EM Induction Survey

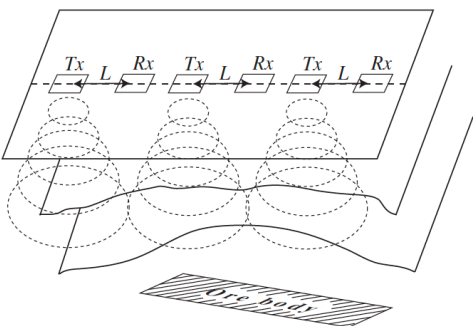


Figure 1.6

Moving transmitter loop (Tx)—receiver loop (Rx) EM induction survey.

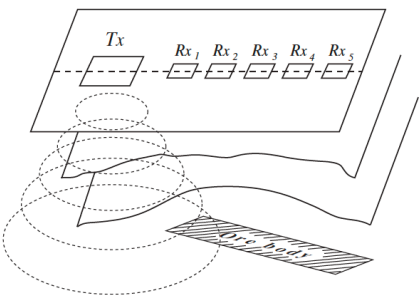


Figure 1.7

Fixed transmitter (Tx) EM induction survey.

There are several approaches to inversion:

Stochastic

Monte Carlo, Markov Chains
Genetic Algorithms
Simulated annealing, etc.
(Bayesian Searches)

Deterministic

Newton Algorithms
Steepest descent
Conjugate Gradients
Quadratic (and Linear) Programming, etc.

All require measuring model fit to the data (i.e., misfit)

Analytical

D+ (1D MT)
Bilayer (1D resistivity)
Ideal body theory in gravity and magnetism

Syllabus review

Term project

Midterm report (due ~Week 9-10):

- Define a forward problem of your choosing, derive the maths, and describe your objective. Generate synthetic forward data from some simple example models. Give presentation in class summarizing your report.

Final report (due end of semester):

- Define one or more inversion strategies to model your forward problem. Design an algorithm, and code it up. Run inversions and assess the model results (e.g., misfit, uncertainties, etc.). Describe and interpret your results. Give presentation in class summarizing the outcomes of your work.

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Homework #1, 2026

This assignment is designed to provide me and your colleagues with information about you, the participants in this class, and what you hope to learn this quarter. At our next class on Wednesday January 14 we'll be discussing your answers to the questions below.

Oral presentation (4-5 minutes) of response to Q1-5 in class on Wednesday January 14. Send me your presentation slides by noon on Wednesday January 14.

1. Please describe your experiences in statistics, linear algebra, optimization, and differential equations. What classes at GTech have you taken on these topics, if any?
2. Have you taken a class in inverse theory before? If yes, was it based on a specific textbook? Please tell us the author and title.
3. Do you have a preferred programming language for doing homework problems? What do you like about it?
4. What kind of data are you interested in? Please provide an example of a geophysical problem you have either encountered during your studies or are interested in solving. In the context of this class, that first requires an idea of how to represent some Earth property and predict observables if you know that property exactly (the forward model). The inverse problem is to recover estimates of the property from a finite number of observations plus noise, and quantify the uncertainty in your result.
5. What do you hope to learn from this class?

Make sure to start reviewing linear algebra!

- Read:
 - Appendix A in *Parameter Estimation*
 - pages 1-19 in Parker supplementary notes

PDFs can be found at: Canvas—>Files—>Reading

We will review this material in class on January 21st

**Arrive early to class on
Wednesday!**